## Internship Overview

# Autonomous Field Navigation of Robot

[ID 022026]

Develop an autonomous navigation stack for a small field robot, using LiDAR and positioning sensors for obstacle detection and safe movement.

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**Objectives**

- Enable autonomous field navigation
- Detect obstacles using LiDAR data
- Fuse localization and perception signals for safer motion

**Tasks**

- Integrate LiDAR, GPS, and Pixhawk interfaces
- Build localization and navigation modules
- Detect obstacles from LiDAR observations
- Plan and execute autonomous paths in the field
- Validate navigation performance on the robot chassis

**Requirements**

- Python programming
- Robotics fundamentals
- ROS basics

**Equipment**

- RPLIDAR C1 LiDAR
- ZED-F9P-01B-01 GPS module
- Holybro Pixhawk 6X flight controller
- UGV Rover ROS 2 Open-source 6 Wheels 4WD

**Deliverables**

- Autonomous navigation prototype with LiDAR-based obstacle detection
- Field validation results

**Work Breakdown**

- 30% hardware integration
- 70% coding

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**Duration:** 2 months

**Payment:** Available in Bizquick

**Supervisor:** Kaleb Granados